

D7 Quantum Calculations

D7.1 An LED 2.3 V across it to make it work. This means that its photons will have an energy of 2.3 eV. Calculate the wavelength of the light emitted.

Complete the question in the table:

	Wavelength /nm	Particle	Momentum /kg m s ⁻¹	Kinetic energy /eV
D7.2	0.50	Elec- tron	(a)	(b)

D7.3 A $^{207}_{82}\text{Pb}$ nucleus is accelerated from rest through a potential of 5.0 MV. How fast is it going? What is its wavelength?

CHAPTER D. WAVES

Answers

D7.1 $\lambda = \frac{c}{f} = \frac{hc}{E} = \frac{hc}{Ve} = 540.49 \text{ nm}$

D7.2

a) $mv = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{5.0 \times 10^{-10}} = 1.326 \times 10^{-24} \text{ kg m s}^{-1}$

b) $E = \frac{(mv)^2}{2m} = \frac{(1.326 \times 10^{-24})^2}{2 \times 9.11 \times 10^{-31}} = 9.662 \times 10^{-19} \text{ J}$
 $E_{\text{eV}} = \frac{E}{e} = 6.0314 \text{ eV}$

D7.3

a) Kinetic energy $E = qV = 82 \times 1.60 \times 10^{-19} \times 5.0 \times 10^6 = 6.56 \times 10^{-11} \text{ J}$.

Speed $v = \sqrt{\frac{2E}{m}} = \sqrt{\frac{2 \times 6.56 \times 10^{-11}}{207 \times 1.66 \times 10^{-27}}} = 1.9552 \times 10^7 \text{ m s}^{-1}$.

b) $\lambda = \frac{h}{mv} = 9.8682 \times 10^{-17} \text{ m}$